

- 4 (a) Table 4.1 shows a sequence of 12 nucleotides in the template strand of a short length of a DNA molecule, the corresponding primary transcript and the four amino acids coded for by the sequence. The table is incomplete.
- (i) Complete Table 4.1 to show the sequence of nucleotides in the primary transcript that would result from transcription of this short length of DNA.

Table 4.1

position of nucleotide	1	2	3	4	5	6	7	8	9	10	11	12
DNA template strand	C	A	C	T	A	C	T	C	C	A	A	C
primary transcript												
amino acid	aa1			aa2			aa3			aa4		

[1]

- (ii) Table 4.2 shows all the possible template strand DNA triplets that code for the amino acids labelled aa1, aa2, aa3 and aa4 in Table 4.1.

Table 4.2

amino acid	DNA triplets
val	CAA, CAG, CAT, CAC
arg	GCA, GCG, GCT, GCC, TCT, TCC
met	TAC
leu	AAT, AAC, GAA, GAG, GAT, GAC

Complete Table 4.3 to identify the four amino acids labelled aa1, aa2, aa3 and aa4 in Table 4.1.

Table 4.3

	aa1	aa2	aa3	aa4
amino acid				

[1]

(iii) One type of gene mutation is caused by the substitution of a DNA nucleotide.

Using the information in Table 4.2, state **and** explain the effect on the final protein structure of a substitution of the nucleotide at position 3 in Table 4.1.

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..... [2]

(iv) A second type of gene mutation is caused by the deletion of a DNA nucleotide.

Using the information in Table 4.2, state **and** explain the effect on the final protein structure of a deletion of the nucleotide at position 3 in Table 4.1.

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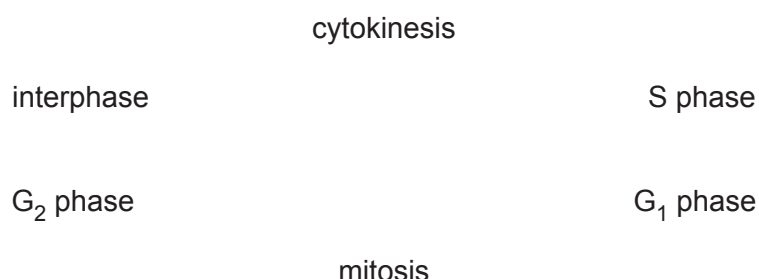
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..... [3]

(b) Replication of nuclear DNA occurs just once in every mitotic cell cycle. Six named events associated with the mitotic cell cycle are listed. The events are **not** listed in any particular order.

Draw a circle around each event where replication of nuclear DNA occurs.



[4]

NC1=NC=NC2=C1N=CN2[C@H]3O[C@@H](COP(=O)(O)OP(=O)(O)OP(=O)(O)O)[C@H](O)[C@H](O)[C@H]3O

State the name of the part of the ATP molecule labelled **A** in Fig. 4.1.

..... [1]

[Total: 13]